



CSE 151B Discussion 2

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7/1/2026



Roll Call

- Not for a grade!



Today's Discussion

- Reminders
- Assignment 1 Overview
- QnA
- Feedback



Reminders

- A0 due **TONIGHT**
- Oral assessments **TOMORROW 7/1**, please show up for the slot you signed up for (CSE Basement 240A or 250A)
 - On the programming portion of A0 - Pytorch tutorial
- Practice exam with Triton Testing Center (aka CBTF) due **THURSDAY 7/2**
- Pre-lecture homeworks due for correctness due **FRIDAY 7/3**
- A0 LPRR due **FRIDAY 7/3**
- Project proposal due **NEXT MONDAY 7/6**



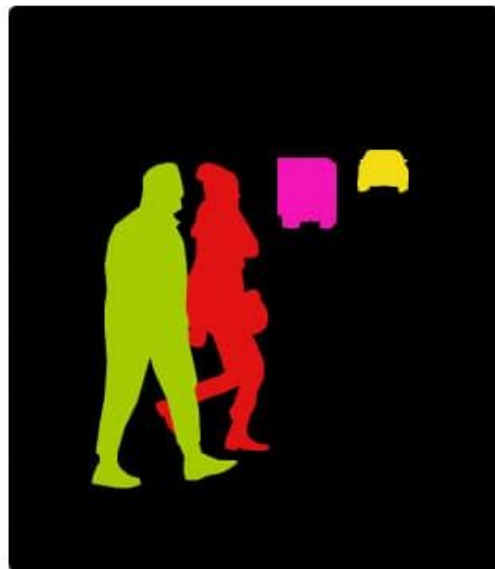
Assignment 1 Overview

- Do not use pytorch for anything other than model saving!
- A1 Part 1 - Neural Network Classification
 - Questions Portion - answers will be submitted on PrairieLearn
 - Programming Portion - code submission details TBA, MCQ answer submissions on PrairieLearn
- A1 Part 2 - Image Segmentation using a Convolutional Neural Network
 - Questions portion - answers will be submitted on PrairieLearn
 - Programming Portion - code submission details TBA

Types of Image Segmentation



**SEMANTIC IMAGE
SEGMENTATION**



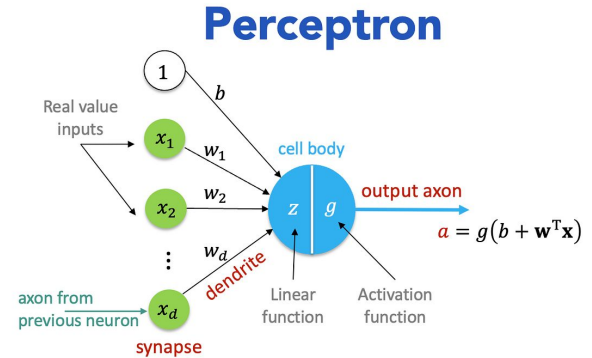
**INSTANCE
SEGMENTATION**



**PANOPTIC
SEGMENTATION**

Assignment 1 Part 1: Q1b Perceptron Demo

- Desmos demo to help visualize the problem
- Walk through the first few steps given to us
- <https://www.desmos.com/calculator/4xnikac9vu>





Assignment 1 Part 1: Programming Outline

- Run a baseline model with softmax
- Check your backprop code by comparing it to a numerical approximation
- Experiment with regularization (L1/L2)
- Experiment with activation functions



Assignment 1 Part 1: Programming Outline Cont.d

- Given to you:
 - main.py
 - Argument parser (--experiment, --data)
 - Config loading (parameters for models)
 - visualize.ipynb
 - Data visualization code
 - util.py
 - Data download and unpacking code
 - Plot generation function
 - configs 4, 5, and 6



Assignment 1 Part 1: Programming Outline Cont.d

- To implement:
 - `neuralnet.py`
 - Implement activation functions and their derivatives in `Activation()` class
 - Implement forward and backward code for individual layers
 - Implement forward, backward, and loss code for the entire neural network
 - `train.py`
 - Implement training loop
 - Implement test set evaluation code
 - config files
 - Implement config 7 and 8

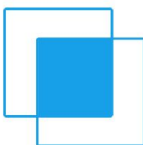


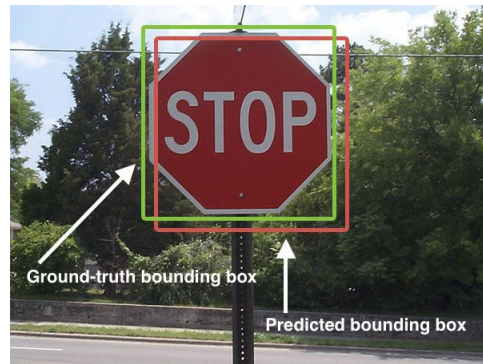
Assignment 1 Part 1: Programming Outline Cont.d

- util.py
 - Implement data normalization
 - Implement onehot encoding code
 - Implement code to calculate number of correct predictions
 - Implement code to append bias
 - Implement code for train and validation split
- gradient.py
 - Implement numerical gradient checking to compare to your backpropagation
- main.py
 - Put together all other components to train, evaluate, save, and load your model.

Assignment 1 Part 2: Programming Outline

- To implement:
 - util.py
 - Implement intersection over union (IoU) metric
 - Implement pixel accuracy metric
 - voc.py
 - make_dataset()
 - Everything else will be implemented in A2!


$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$



Any Questions, Comments, or Concerns?



Feedback Form

